Beyond Linear Constant Coefficients: A Ladder of Extensions for Reading Equations from Data

Flavio Pricoco*

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Abstract. The factor pipeline reads a linear, constant-coefficient, scalar equation in one space and one time variable. Real equations are often outside this class: they are nonlinear, have a potential, depend on time, form a system, or are nonlocal. This paper maps what can be said beyond the class, level by level. We organise the extensions as a ladder of eleven levels and give an extended algorithm with six steps, each stating what it takes, what it returns, which results justify it and how it fails. Every output carries two labels: a completeness label (is the right law inside the level searched?) and a correctness label (exact, convergent with a bound, no guarantee, or only in the limit). The main findings are positive and negative. Positive: for a nonlinear equation, an interaction term can be defined, expanded in time and written in closed form before a shock; with a declared dictionary, identification of Burgers' equation with N shocks is complete; for a variable potential $V(x)$ every nonzero solution in a regular class identifies V almost everywhere; for first-order systems such as Dirac, a point-wise linear system of rank 8 identifies the law. Negative: finite exact data never certify a fit, the annihilator is discontinuous everywhere so no Lipschitz bound exists for the discrete steps, no time-independent encoder exists for a generic $V(t, x)$, and a bounded transformation loses about four orders of magnitude in its transported certificates. Four worked examples (Dirac, Pöschl–Teller, Stark, and a dictionary on the heat equation) are checked symbolically. We end with the minimal route for extending the code and a list of open questions.

Key words. equation identification, variable coefficients, nonlinear dictionary, systems, annihilator ideal, certificates

MSC codes. 35R30, 35E20, 35Q41, 68W30

1 Introduction

Three equations show what changes when the pipeline's class is left.

$$u_t + uu_x = \nu u_{xx} \text{ (Burgers)}, \qquad iu_t = -u_{xx} + q(x)\,u \text{ (potential)}, \qquad iu_t = -u_{xx} + t\,x\,u \text{ (Stark)}. \tag{1}$$

The first is nonlinear: solutions do not form a vector space, and there is no polynomial symbol whose zero set carries the modes. The second has a coefficient that depends on x: Fourier modes are no longer eigenfunctions, and an exact formula such as the one for $q = -2\,\mathrm{sech}^2\,x$ mixes a bound state with a scattering state. The third depends on t as well, so there is no autonomous flow. For each, the questions of the base problem return in a new form. Is the law determined by one solution? Can a solution be split by term or by factor? What can an algorithm promise from noisy samples?

*flaflapr@gmail.com

This paper answers these questions where answers exist and says where they do not. It has three parts. Section 2 fixes the vocabulary: five meanings of "identify" and four classes of guarantee. Section 3 states the extended algorithm over a ladder of levels. Section 4 reviews each direction of extension with its proven results and its limits. Section 5 works four examples. Section 6 describes the smallest sequence of steps by which the code can be extended, and Section 7 lists what is open.

Scope and status The proofs are in the companion paper on the analytical branch. Every statement is quoted with its status: *proved* (a complete proof exists, internally audited), *known* (a published result with a verified source), *conditional* (depends on a hypothesis or an unverified citation, named), *open*, or *false* (a counterexample is recorded). The audits were done by independent Claude reviewers, not human referees. Nothing here has been implemented in the pipeline: the extended algorithm is a specification, and its worked examples are symbolic checks, not benchmarks on noisy data.

2 Vocabulary: what "identify" means and what a guarantee is

2.1 Five senses of identification

A solution can determine an equation in different ways, and conflating them gives wrong claims. We write them explicitly.

sense	meaning
ID-ann	the annihilator ideal of u is principal and generated by the law. Never true for an exponential-polynomial solution.
ID-ker	the linear system for the coefficients has trivial kernel inside a declared class, *provided* a law of the class exists (hypothesis H-ex).
ID-res	the coefficient is read from the residual map, for example $V = (iu_t + u_{xx})/u$ where $u \neq 0$.
ID-fam	identification modulo a family: $V + c(t)$ for data known up to a global phase, or a gauge class.
ID-lim	identification in a limit of increasing level, never certified at finite level.

The distinction matters: a solution $u = e^{-t}\cos x$ of the heat equation also lies in the kernel of $\partial_t + 1$ and $\partial_x^2 + 1$, so its annihilator is not generated by the heat operator, and ID-ann fails; yet ID-ker in the class $u_t = a(\partial_x)u$ of bounded degree can succeed.

2.2 Data models

model	datum	senses allowed
DM1	finite exponential-polynomial formula	ID-ker in a finite class; never ID-ann
M3-hol	exact holonomic closed form (Bessel, Airy, sech)	ID-ann, ID-res, ID-ker
DM2	jet at one point	ID-lim only; no finite certificate
DM3	u known exactly on Ω	ID-ann, ID-res, ID-ker, ID-fam
DM-φ	u up to a global phase $e^{i\varphi(t)}$	ID-fam (V modulo functions of t)
DM-$\delta(Y)$	$\tilde{u}$ with $\|\tilde{u}-u\|_Y \le \delta$, norm declared	no identification; stability bounds

A noise model without a declared norm Y is undefined: in C^1 there is no control of a potential, since $u = 1$ and $u_\delta = 1 + \delta^2 \sin(x/\delta)$ are $O(\delta)$-close in C^1 while V_δ reaches $1/(1-\delta^2)$.

2.3 Four classes of guarantee, two layers

Each step of the extended pipeline has one of four contracts: **(A)** finite and exact; **(B)** convergent with an explicit error ε_i; **(C)** no guarantee; **(Lim)** in the limit, without a certificate. The pipeline has two layers. *Conditioning* (data model, level, identification) produces discrete outputs: a model, a level, an ideal, a class. The annihilator is discontinuous at every point (both the principal and the non-principal ideals are dense), so there is no Lipschitz constant and no error bound applies to these steps. *Conditional correctness* (misfit certificate, encoder, decomposition) has metric outputs; at the declared level and class, and if each step is (A) or (B) and Lipschitz, errors compose as $d(\tilde{x}_k, x_k) \le \sum_i (\prod_{j>i} L_j)\varepsilon_i$. A step of class (C) or (Lim) makes everything after it (C) or (Lim). Therefore each output has *two* labels: completeness (is the right law in the level searched? almost always (C), since the level is chosen by a heuristic and no a priori criterion exists) and conditional correctness. The output labels never say "law identified" from finite numerical data: FIT-COMPATIBLE, MISFIT, UNDECIDED, ID-KER and, only with an exact symbolic proof, a sense-tagged `IDENTIFIED[sense, model]`.

3 The ladder and the extended algorithm

3.1 Levels

The dictionary of admissible atoms is filtered by height. The levels that the algorithm can be asked to work at are:

level	class	proved tools
L0–L2	scalar, constant coefficients, monic in τ, increasing degree	annihilator criterion, spectral form
L3	non-monic or degree ≥ 2 in τ	Gröbner-type reduction
L4	polynomial nonlinear	interaction term, dictionary-relative ideal
L5	memory and delay, via an augmented system	delay as transport
L6	constant-coefficient systems (Dirac, Lamé, Maxwell)	module theory, rank criteria
L7	coefficients depending on x	residual map, encoders
L8	coefficients depending on (t, x) (Stark, Floquet)	residual map, gauge transforms
L9	nonlocal (fractional, Fourier multipliers)	non-identifiability theorems
L10	general dictionary with misfit test and stopping rule	Hahn–Banach residual, Gram determinants

The level is chosen by an ascending search with the stopping rule R5; the search is heuristic, so completeness is (C). If the budget is exhausted the output is UNDECIDED.

3.2 The six steps

1. **Data model** (conditioning). Fix the model of Section 2, the coefficient ring, the domain and the budget. Proved limits: no finite jet at one point certifies the annihilator; for exponential-polynomial data, the jet of order $2d-1$ determines it; a black-box identification without extra data is impossible.
2. **Level** (conditioning). Pick the level n and a finite set of atoms $\mathcal{D}_n$. No countable dictionary is exhaustive (the space of smooth functions is not separable), a delay is at distance ≥ 1 from the local dictionary, and no learner identifies the annihilator on the union of principal ideals and the zero ideal.
3. **Identification** (conditioning). Decide ID-ann, ID-res or relative ID-ker at level n by the criteria of the direction in question (Section 4).
4. **Misfit certificate or best fit** (correctness). Compute the distance r_n of $D_0 u$ to the span of the atoms as a *least-squares residual*, not as a ratio of Gram determinants, which is numerically singular when $D_0 u$ lies in the span. A misfit is certified if $r_n > 0$ is certified; it is class (A) only with a rigorous enclosure of r_n and class (B) with floating-point quadrature. A fit is class (B) at the declared level, with bound $|c-\tilde{c}| \leq 2\eta/\sqrt{\lambda_{\min}}$, and *is never certifiable from finite exact data.*
5. **Encoder** (correctness). Translate a variable-coefficient problem to a constant-coefficient one *for a known potential*; the encoder is a translation, not a source of information. Bounded (Volterra) translations preserve certificates up to a constant $C_W = 1+a^2 Q e^{a\sqrt{Q}}$ ($Q = \|q\|_\infty$); Darboux translations are unbounded, have a nontrivial kernel and are class (C).
6. **Decomposition** (correctness). Split into components by the factors of the law and quantify the interaction term between terms of a factor.

4 The extensions, direction by direction

Nonlinear For two terms of an equation with flows Φ_n, the interaction term is the part of the solution that is not a sum of single-term pieces; its time jet has first nonzero coefficient $\frac{1}{2}\sum_n T_n'[T_m]$ at order t^2, e.g. $-\nu(u_0 u_0''' + 2u_0' u_0'')t^2$ for Burgers. Before the shock a closed form holds at first order in ν with remainder $\nu^2\varepsilon^2 t^3$; at the shock time the solution blows up in a way

Table 1: Direction, what is proved, and what blocks the next step. Status words as in Section 1.

direction	proved	open or impossible
Nonlinear, polynomial	interaction term E_{int} defined by the flows of single terms; jet in t; closed form before the shock at first order; blow-up time; identification relative to a dictionary	several shocks; periodic Korteweg–de Vries; non-uniform estimates in t
Nonlinear dictionary	misfit certificates; fit not certifiable from finite data; dictionary-relative rules R1–R6; space of u with a law in the dictionary is meagre	division with variable rank; some parked sub-cases
Variable $V(x)$	every nonzero u in a regular class identifies V a.e.; Wei–Norman closure finite iff $\deg V \leq 2$; structured identification for D-finite u	multiplications; non-monic symbols; Weyl algebra for PDE
Non-autonomous $V(t,x)$	identifies iff the zero set of u has empty interior (continuous V) or measure zero (L^∞); no t-independent encoder for a generic V; data up to phase give $V + c(t)$	the class $W(t)$ for reading the third data model
Encoders	bounded Volterra translation; Darboux chains; transported certificates	tighter bound than C_W; Lipschitz constant of $q \mapsto W_q$
Systems	existence of an identifying u for $\det P \neq 0$; Dirac, Lamé; magnetic family only up to gauge	Dirac with complex mass
Higher dimension	ambiguity is infinite unless the factors are comaximal; split and decomposition	sum closure for $N \geq 3$; non-Lipschitz domains
Convergence, noise	stability of the misfit residual; discrepancy rule selects the right degree; locality trichotomy	threshold δ_0 not bounded below by data; unbounded V
Nonlocal	indistinguishability from a local law for finite data; no finite degree in the limit	example with infinitely many modes

set by the data, and beyond it the jumps of the additive part are independent of ν and forced by the definition. No finite factorisation of the flows exists for any datum. Relative to a dictionary, identification reduces exactly to the sequence $0 \to [F] \cap V \to I(u) \cap V \to J_u \cap \psi(V) \to 0$: for Burgers with N shocks the criterion is complete in the evolutionary dictionary (the law is identified iff $d \le N$); for N-soliton Korteweg–de Vries the ideal is principal, $\deg \Pi_2 = 8$ generically with an exact certificate, and $\deg \Pi_N = (N+2)2^{N-1}$. The literal form of "one solution identifies the dictionary law" is false in complete dictionaries and must be stated modulo $[F] \cap V$.

Dictionary and misfit A misfit is an open condition with a finite certificate; a fit is a closed condition with empty interior. Hence no finite exact datum certifies a fit, which is why the algorithm outputs FIT-COMPATIBLE with a bound and never "identified".

Variable coefficients With the coefficient field declared, $V(x)$ is identified a.e. by any nonzero u in a regular class; for $V(t,x)$ the condition moves to the zero set of u. Counterexamples are recorded: a Tychonoff-type series breaks the statement beyond the silence bands. Finite closure of the Wei–Norman expansion holds iff $\deg V \le 2$ (Stark and the oscillator), so beyond quadratic potentials the exact flow is not a finite product of simple flows.

Systems For a 2×2 system the existence of an identifying solution follows from $\det P \neq 0$ in two or more variables; for Lamé it needs $\rho \neq 0, \mu \neq 0, \lambda + \mu \neq 0, \lambda + 2\mu \neq 0$ and fails at $\lambda = -2\mu$. For Dirac in $1+1$, with $m \neq 0$ and the matrix basis fixed, the Hamiltonian class $i\psi_t = A\psi_x + B\psi$ has eight unknowns and the pointwise system has rank 8 for a non-exponential datum, and also for an exponential datum with at least four generic modes; one plane wave gives rank 2. With a magnetic field the solution set is an affine family, never unique: only identification modulo gauge.

Higher dimension and nonlocal In 2+1 and 3+1 the ambiguity is infinite unless the factors are comaximal, so the factor split needs an extra condition. For nonlocal laws finite data cannot distinguish $u_t = -|D|u$ from a local law (every finite-mode datum satisfies a local law by symbol interpolation), and in the limit the multiplier degree $m(\delta) \to \infty$ means the pipeline can only report UNDECIDED with "no finite degree".

5 Four worked examples

All four are in `examples/`; each prints its checks and the number that pass. They are symbolic or high-precision verifications, not proofs.

Dirac, 1+1 (7/7 checks) With $H = -i\sigma_x\partial_x + m\sigma_z$, $m = 1$, $(i\partial_t - H)(i\partial_t + H) = -(\tau^2 - \xi^2 + m^2)I$, so every component solves Klein–Gordon. Take $\psi = (i\partial_t + H)(fe_1)$ with $f = J_0(m\sqrt{t^2 - x^2})$; the residual is below 10^{-20} at 40 digits. The second component is not an exponential-polynomial, so the annihilator premise holds; in the Hamiltonian class the pointwise system has rank 8 ($\sigma_{\min} \approx 4.3 \times 10^{-2}$) and the true law solves it. With one plane wave the rank is $2 < 8$. *Lesson:* failure of ID-ker depends on the number of modes, not on the datum being exponential-polynomial; the criterion $\psi_2 \notin \mathrm{EP}$ is for ID-ann.

Pöschl–Teller (11/11) $q = -2\operatorname{sech}^2 x$, $u = e^{it}\operatorname{sech} x + e^{-ik^2t}(ik - \tanh x)e^{ikx}$, $k = \frac{3}{2}$. The residual returns $V = q$: ID-res. It is not ID-ann, since u is annihilated by $(L_q + 1)(L_q - k^2)$ and $(i\partial_t + 1)(i\partial_t - k^2)$. Data up to phase gives $q - \varphi'(t)$ (ID-fam). The Darboux operator $A = \partial + \tanh x$ intertwines $AL_q = L_0A$, sends u to a free solution, kills the bound state ($\ker A = \mathbb{C}\operatorname{sech}$) and sends the scattering part to a plane wave, an exponential-polynomial datum. A is unbounded, so C_W does not apply: class (C), not an instance of the bounded translation.

Stark (5/5) $V = \beta(t)x$, $\beta = t$. With $B = t^2/2$, $s = -t^3/3$, $g = -t^5/20$, $u = e^{i(-Bx+g)}w(t, x-s)$ and w a free Gaussian solves the equation (direct check below 10^{-30}). u has no zeros, so $V(t, x)$ is identified by the residual; with unknown phase only V modulo functions of t. The Avron–Herbst transformation gives class (A); the Magnus expansion is (C), because $\int \|A_I\| = \infty$.

Dictionary on the heat equation (12/12) $u = (t+1)^{-1/2}e^{-x^2/(4(t+1))}$ on $(0, 1) \times (-1, 1)$, $Y = L^2$, quadrature at 25 digits. At the affine level (atoms $1, \partial_x$) $r_1 = 0.0778 > 0$: a misfit, class (B) because the quadrature has no rigorous enclosure. At level 2 (atoms $1, \partial_x, \partial_x^2, \partial_t\partial_x, \partial_t^2$), $r_2 \approx 4 \times 10^{-33}$ with minimiser $\partial_t - \partial_x^2$, and the homogeneous Gram matrix has one null eigenvalue (1.25×10^{-28}; the next is 1.71×10^{-4}): output FIT-COMPATIBLE with bound, not "identified". Under a perturbation $\varepsilon \cos 3x$ the stability bound is vacuous at $\varepsilon = 10^{-3}$ ($2\delta/\rho = 1.43 > 1$) and non-trivial at 10^{-4} (0.143). Finally $u = t + \sin x$ in the class $u_t = au_{xx} + bu$ has $\ker \Phi_u = 0$ but no law exists: ID-ker without H-ex identifies nothing. A nonlocal example with infinitely many modes that identifies a nonlocal atom is still missing.

6 A minimal route for extending the code

The pipeline refuses before it extends. Its validators already compare residuals per cell and per time window: a nonlinear equation shows as amplitude-dependent coupling, a variable coefficient as drift of the dispersion relation across cells, two laws as two clusters. The minimal route is therefore:

1. **Detect and refuse.** Report MISFIT or UNDECIDED with the reason (nonlinear, variable coefficient, fractional, two laws). Already specified; the data generator `benchmarks/out_of_class_inject.py` builds such inputs (nonlinear, variable coefficient, fractional, two laws, coloured noise, heteroscedastic noise, aliasing) at scale-free strengths.
2. **Frozen-coefficient cells** for slowly varying $V(x)$: fit each cell with the constant-coefficient pipeline and read the drift (the domain quadtree and aggregate code exist).
3. **Structured identification** (the residual map) with a declared coefficient class.
4. **Nonlinear only through a declared dictionary**, never free-form, with the misfit certificate of step 4.

An evaluation found that a learned detector flags none of the injected nonlinear, variable-coefficient or two-law failures; the first step therefore needs hand-built tests with a theoretical null (the residual per bin and the smooth-structure test), not a learned score.

7 Open questions and honest limits

1. Robustness under bounded noise DM-$\delta(Y)$ for variable coefficients.
2. An example of infinitely many modes that identifies a nonlocal atom, and non-certifiability of fits for nonlocal atoms.
3. Cost of the dictionary search per level; an audit of the polynomial-cost elimination algorithm; a rigorous enclosure for the misfit residual (to move it from class (B) to (A)).
4. A bound for the bounded translation tighter than C_W: at $a\sqrt{Q} = 5$, $C_W^2 \approx 1.4 \times 10^7$ while $\cosh^2(a\sqrt{Q}) \approx 5.5 \times 10^3$, so transported certificates may lose about four orders of magnitude; and an explicit Lipschitz constant of $q \mapsto W_q$ for an estimated potential.
5. Non-monic symbols (BBM type): the annihilator is proved principal only for symbols monic in τ.
6. Higher dimension: sum closure for $N \geq 3$ and non-Lipschitz domains.

Honest limits The overall novelty is modest: many building blocks are classical (Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering). The contribution is the package of characterisations, negative theorems and counterexamples with verified statuses. The reviewers were independent Claude agents, not human referees. The source bibliography has 112 entries, of which some load-bearing theorem numbers remain unchecked; a few results are labelled conditional on them.

8 Conclusion

Beyond the linear constant-coefficient class the algorithm can promise little at finite data, and the paper says what. What can be proved is structural: which solutions identify which laws, in which sense, and with which counterexamples. The pipeline should therefore grow by refusing correctly first, then by adding structured classes one at a time, each with its sense of identification and its contract label.